

ARTIFICIAL INTELLIGENCE (CSA1703)

Assessment Tool – 1

CO1: Intelligent Agents and Uninformed Search Methods

Question 1: Water Jug Problem

Problem

Two jugs of capacities **4 gallons** and **3 gallons** are given. Obtain exactly **2 gallons** in the 4-gallon jug.

Solution

Step	Action	4-Gallon Jug	3-Gallon Jug
1	Fill the 3-gallon jug	0	3
2	Pour into the 4-gallon jug	3	0
3	Fill the 3-gallon jug again	3	3
4	Pour into the 4-gallon jug until it becomes full	4	2
5	Empty the 4-gallon jug	0	2
6	Transfer the remaining 2 gallons into the 4-gallon jug	2	0

Result

Hence, exactly **2 gallons** of water is obtained in the **4-gallon jug**.

Question 2: Mars Rover Intelligent Agent

i) Percepts

- Camera images
- Rock and soil samples
- Temperature
- Atmospheric pressure
- Chemical sensor readings
- Terrain information

ii) Operating Environment

- Partially Observable
- Deterministic
- Sequential

- Dynamic
- Continuous
- Single Agent

iii) Actions

- Move forward/backward
- Turn left/right
- Capture images
- Drill rocks
- Collect soil samples
- Analyze samples
- Send collected data to Earth

iv) Performance Measure

- Accuracy of scientific data
- Number of samples collected
- Safe navigation
- Efficient energy usage
- Successful mission completion

v) Suitable Agent Architecture

Model-Based Goal-Based Agent

Reason:

The Mars Rover maintains knowledge of its environment, plans its path, avoids obstacles, and achieves mission goals autonomously.

Question 3: 8-Queens Problem

Problem Formulation

Initial State

- Empty chessboard.

State Space

- All possible arrangements of eight queens on the chessboard.

Operators

- Place one queen in a safe position.

Goal State

- No two queens attack each other horizontally, vertically, or diagonally.

Path Cost

- Each queen placement has a cost of one.

Search Strategy

- Backtracking using Depth-First Search (DFS).
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Question 4: OLA Cab Booking**Type of Problem-Solving Agent****Goal-Based Agent****Reason**

The customer's goal is to reach the destination by selecting the most suitable cab based on comfort, availability, and fare.

Algorithm (Pseudo Code)

START

Input Pickup Location

Input Destination

Input Preferred Cab Type

Display Available Cabs

IF Preferred Cab is Available THEN

 Select Preferred Cab

ELSE

 Select Nearest Available Cab

ENDIF

Calculate Fare

Confirm Booking

Assign Driver

Display Driver Details

STOP

Question 5: Uniform Cost Search (UCS)**Given Graph**

Path Cost

S → A 1

Path Cost

$S \rightarrow G$ 12

$A \rightarrow B$ 3

$A \rightarrow C$ 1

$B \rightarrow D$ 3

$C \rightarrow D$ 1

$C \rightarrow G$ 2

$D \rightarrow G$ 3

UCS Expansion

Expanded Path	Total Cost
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S	0
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$S \rightarrow A$	1
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$S \rightarrow A \rightarrow C$	2
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$S \rightarrow A \rightarrow C \rightarrow G$	4
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$S \rightarrow A \rightarrow C \rightarrow D \rightarrow G$	6
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$S \rightarrow A \rightarrow B \rightarrow D \rightarrow G$	10
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$S \rightarrow G$	12
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Least Cost Path

$S \rightarrow A \rightarrow C \rightarrow G$

Total Cost

4

Conclusion

Uniform Cost Search expands the node with the minimum cumulative cost and guarantees the optimal solution when all path costs are non-negative.

This format is suitable for writing directly as an **engineering assignment**, with proper headings, subheadings, tables, and conclusions.